(b) Interchange Saxena and Kumar as well as David and Jain
(c) Interchange David and Kumar as well as Saxena and Jain
(d) It is not possible for all the four persons to have been posted at all the four places

Ans. (c) : When David and Kumar interchange their positions and Saxena and Jain interchange their positions, the four people will be able to be stationed at all four places.

139. In a group of travelling in a bus, 6 persons can speak Tamil, 15 can speak Hindi and 6 can speak Gujarati. In that group none can speak any other language. If 2 persons in the group can speak two languages and one person can speak all the three languages, then how many persons are there in the group?

(a) 21 (b) 22
(c) 23 (d) 24

Ans. (d) : Total number of Hindi, Tamil and Gujarati speaking people = 15 + 6 + 6 = 27
Two persons of this group can speak 2 languages while one can speak 3 languages.
Whereas restriction is that no person of this group can speak any other language.
Hence, number of persons = 27 – 3 = 24.

140. Examine the following statements regarding a set of balls:

1. All balls are black
2. All balls are white
3. Only some balls are black
4. No balls are black

Assuming that the balls can only be black or white, which two of the statements given above can both be true, but cannot both be false? Choose the correct answer from the codes given below:

(a) 1 and 4 (b) 1 and 3
(c) 2 and 3 (d) 2 and 4

Ans. (a) : As per the statement given above the condition for ball being only black or only be white is.
(i) All balls are black which means there is no white balls and all are black balls.
(iv) No balls are black means all balls are white.

141. Daily weather map showing isobars is an example of:

(a) Choropleth map
(b) Isopleth map
(c) Chorochromatic map
(d) Choroschematic map

Ans. (b) : Isopleths is a line on map connecting places registering the same amount or ratio of some geographical phenomenon or phenomena. The example of isopleths is daily weather map showing Isobars.

142. A boat which has a speed of 5 km/hr in still water crosses a river of width 1 km along the shortes possible path in 15 minutes. The velocity of the river water in km/hr is:

(a) 1 (b) 3
(c) 4 (d) $\sqrt{41}$

Ans. (a) : The speed of boat is 5 km/hr in still water = U
The velocity of river water = V
As, boat is going against the stream of river water– [U –V]

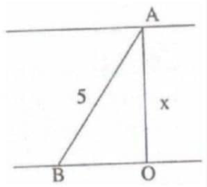
$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

$$[U - V] = \frac{1\text{km}}{15\,\text{minute}}$$

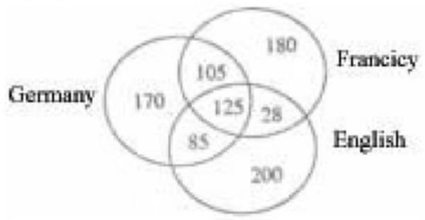
$$[5 - V] = \frac{1\text{km}}{\frac{15}{60}\text{hour}}$$

$$5 - V = \frac{60}{15}$$

$5 - V = 4\text{km/hour}$
$V = (5 - 4)\text{km/hr}$
$V = 1\text{km/hr}$



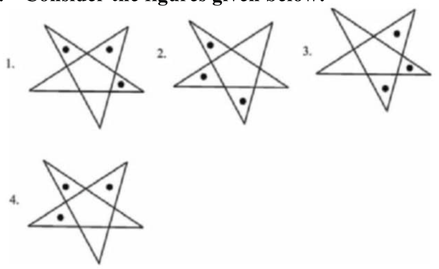
143. A survey was conducted on a sample of 1000 persons with reference to their knowledge of English. French and German. The results of the survey are presented in the given Benn diagram. The ratio of the number of the persons who do not know any of the three languages to those who know all the three languages is



(a) $\frac{1}{27}$ (b) $\frac{1}{25}$
(c) $\frac{1}{550}$ (d) $\frac{175}{1000}$

Ans. (b) : Total number of persons who know only English or French or German = 200+ 180 +170 = 550
The number of persons who know any two languages = 105 + 85 +78 = 268
The number of persons who know all the three languages. = 175
∴ The number of persons who know any of the languages = 550+268+175 = 993
Then :- The number of persons who do not know any of the language = 1000 – 993 = 7
Then the ratio of know not any languages $= \frac{7}{175} = \frac{1}{25}$
know all language.

144. Consider the figures given below:



In the given sequence of figures, the figure that would logically appear at SI no. 5 is

(a) 
(b) 
(c) 
(d) 

Ans. (a) : According to above figure analysis the fifth number figure is -



Hence option (a) is correct answer.

145. The sequencing of the entire genome (the totality of all genes) of an organism was completed in 1996. The organism was:

(a) Albino mouse (b) Yeast
(c) Human being (d) Plasmodium vivax

Ans. (c) : The sequencing of all the entire genome (the totality of all genes) of an organism was completed in 1996. The organism was none other than human being. Human whole genome genome sequencing (WGS) offers the most detailed view into our genetic code. WGC has the ability to evaluate every base in the genone and navigate the complexity of genomic variants that make us unique.

146. Which one of the following is NOT a principle of "Panch-sheel"?

(a) Non-alignment
(b) Peaceful Co-existence
(c) Mutual respect for each other's territorial integrity and sovereignty
(d) Mutual non-interference in each other's internal affairs

Ans. (a) : The five principles of peaceful coexistence, also known as panchsheel, were mentioned in the preamble of the Sino Indian Agreement, 1954.
The five principles are as follows:
(i) Mutual respect for each other's territorial integrity and sovereignty.
(ii) Mutual non-aggression.
(iii) Mutual non-interference in each other's internal affairs.
(iv) Equality and mutual benefit.
(v) Peaceful co-existing.

147. During a flight from Delhi to Tokyo the following are the landing airports:

1. Hongkong 2. Hanoi
3. Taipei 4. Bangkok

The correct sequence of the landing at these airports during an onward journey is

(a) 1, 2, 3, 4 (b) 4, 2, 1, 3
(c) 3, 4, 1, 2 (d) 4, 1, 2, 3

Ans. (b) : The flight landing airports while travelling from Delhi to Tokyo are as follow.
Bangkok → Hong kong → Hanoi → Taipei
So, correct sequence will be 4, 2, 1, 3.

148. Antigen is a substance which:

(a) Lowers body temperature
(b) Destroys harmful bacteria
(c) Triggers the immune system
(d) Iss used as an antidote to poison

Ans. (c) : The term antigen originally referred to a substance that is an antibody generator. The presence of antigens in the body may trigger an immune response. Antigens can be proteins, peptides, polysaccharides and lipids.

149. The number of times in a day the Hour-hand and the Minute-hand of a clock are at right angles is:

(a) 44 (b) 48
(c) 24 (d) 12

Ans. (a) : (i) Minute hand moves 360 degrees in 60 minute.
So, angle is let say '6t'[t is number of minute past midnight]
(ii) The hour hand moves 30 degree in 60 minute.
So, angle is '0.5t'
The first time they make a 90° degree angle is when the minute hand has moved 90 degree further than hour hand.
So
$6t = 0.5t + 90$ ……..(i)

$$t = \frac{.90}{5.5} \quad \boxed{t = 16\frac{4}{11}}$$

Next time when the minute hand has gained another 180° on the hour hand and is 90° behind it.
$6t = 0.5t + 270$

$$\boxed{t = \frac{270}{5.5} = 49\frac{1}{11}} \quad \text{……..(ii)}$$